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链路跟踪

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版本变更说明

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前言

本文档是产品的用户使用手册之一。

阅读前注意事项

通过阅读本文档，您确认并同意自行承担因未具备必要专业背景 and 知识而导致的任何风险或后果。在使用本文档中提供的信息和指南时，请始终谨慎，并在必要时寻求专业人士建议和指导。

- 适用对象

本文档主要适用于使用本产品的系统管理员阅读，部分内容同样适用于基于本产品进行应用开发或应用部署的人员阅读。

- 专业背景

本文内容可能涉及到操作系统、服务器硬件、网络等相关领域的知识。请确保您具备相关背景 and 知识，以便更好地理解 and 应用本手册的内容。

- 技能要求

为了能够充分理解 and 应用本文档的内容，建议您具备如下技能：

- 掌握 Linux 系统基本操作；
- 熟悉数据库（MySQL 等）。
- 安装产品。

- 术语和概念

本文档可能使用一些专业术语和概念。请确保您熟悉这些术语和概念，或者有能力查阅相关资料以便进一步理解。

- 实践经验

为了最大程度地受益于本文档，建议您具备一定的实践经验。这将帮助您更好地应用文档中的操作指南 and 建议。

请注意，本文档不适用于没有相关专业背景 and 知识的用户。

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用户手册集

为您提供的用户使用手册集包含如下文档：

编号	手册名称	说明
1	001_TongESB_V7.1.3.0 使用手册	提供TongESB安装及使用说明。
2	002_TongESB_V7.1.3.0 数据库服务化操作指导	详细介绍数据库服务化中创建、配置、部署、调用数据服务的详细操作。

编号	手册名称	说明
3	003_TongESB_V7.1.3.0 数据映射操作指导	详细介绍数据映射的详细操作。
4	004_TongESB_V7.1.3.0 EDI操作指导	详细介绍EDI的详细操作。
5	005_TongESB_V7.1.3.0 动态路由语法规则手册	提供API动态路由语法规则说明。
6	006_TongESB_V7.1.3.0 链路跟踪	提供链路跟踪的安装及配置。
7	007_TongESB_V7.1.3.0 安全加固配置说明	提供安全加固配置说明。
8	008_TongESB_V7.1.3.0 常见问题	提供常见问题的解决方案。

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- 您的联系方式
- 操作系统及其版本
- 产品版本号
- 日志等错误的详细信息

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1. 搭建 Skywalking 环境

默认情况下，系统并未开启 **skywalking** 菜单和 **skywalking agent** 上报功能。

若需要开启如上功能，则需要安装 **skywalking** 和 **skywalking agent**。

Skywalking 版本及下载连接，如下所示。

- [Skywalking apm 9.3.0](#)
- [Skywalking java agent 8.16.0](#)
- [Elasticsearch Windows 7.8.0](#)
- [Elasticsearch Linux 7.8.0](#)

1.1. Windows平台

1.1.1. 安装并启动 Elasticsearch

安装 Skywalking 之前，需要先按照 ElasticSearch 作存储。

如下以“Elasticsearch Windows 7.8.0”为例说明。

1. 下载 [Elasticsearch Windows 7.8.0](#) 。
2. 解压下载的 Elasticsearch 安装包。

直接解压即可，路径中尽量不包含中文和空格。

3. 进入 Elasticsearch 安装目录。
4. 执行如下命令，启动 Elasticsearch。

```
elasticsearch-7.8.0\bin\elasticsearch.bat
```

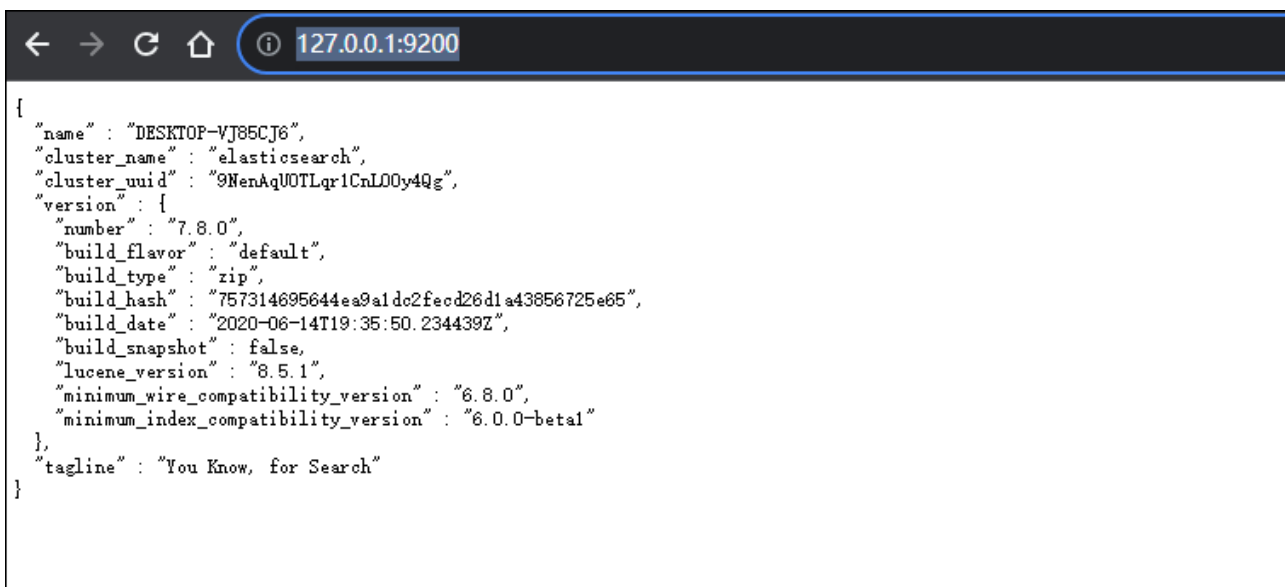
5. 在浏览器中，访问如下地址，检查 Elasticsearch 是否启动成功。

```
http://IP:9200/
```

说明：

IP：表示安装 Elasticsearch 的主机的 IP 地址。

若访问成功，则如下图所示。



```
{
  "name" : "DESKTOP-VJ85CJ6",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "9NenAqUOTLqr1CnL00y4Qg",
  "version" : {
    "number" : "7.8.0",
    "build_flavor" : "default",
    "build_type" : "zip",
    "build_hash" : "757314695644ea9a1dc2fec26d1a43856725e65",
    "build_date" : "2020-06-14T19:35:50.234439Z",
    "build_snapshot" : false,
    "lucene_version" : "8.5.1",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
```

1.1.2. 安装并启动 skywalking

本章节介绍如何安装并启动 skywalking。

注意事项

Skywalking 安装目录命名不要使用空格，使用空格容易导致启动失败。

前置条件

已获取产品工具包中的“skywalking-webapp.jar”。

“skywalking-webapp.jar”针对管理中心进行了部分定制化。

操作步骤

1. 下载 [skywalking apm 9.3.0](#)。
2. 解压下载的 skywalking 安装包。
3. 进入“apache-skywalking-apm-bin\config”目录。
4. 打开“application.yml”文件。
5. 修改如下配置信息。

- 将“storage:selector”修改为“elasticsearch”。

```
storage:
  selector: ${SW_STORAGE:elasticsearch}

elasticsearch:
  namespace: ${SW_NAMESPACE:"elasticsearch"}
  clusterNodes: ${SW_STORAGE_ES_CLUSTER_NODES:localhost:9200}
  protocol: ${SW_STORAGE_ES_HTTP_PROTOCOL:"http"}
  connectTimeout: ${SW_STORAGE_ES_CONNECT_TIMEOUT:3000}
  socketTimeout: ${SW_STORAGE_ES_SOCKET_TIMEOUT:30000}
  responseTimeout: ${SW_STORAGE_ES_RESPONSE_TIMEOUT:15000}
  numHttpThreadPool: ${SW_STORAGE_ES_NUM_HTTP_CLIENT_THREAD:0}
  user: ${SW_ES_USER:""}
  password: ${SW_ES_PASSWORD:""}
  trustStorePath: ${SW_STORAGE_ES_SSL_JKS_PATH:""}
  trustStorePass: ${SW_STORAGE_ES_SSL_JKS_PASS:""}
  secretManagementFile: ${SW_ES_SECRETS_MANAGEMENT_FILE:""} # Secrets management file in the properties format includes the username, password, which are managed by 3rd party tool.
  dayStep: ${SW_STORAGE_DAY_STEP:1} # Represent the number of days in the one minute/hour/day index.
  indexShardsNumber: ${SW_STORAGE_ES_INDEX_SHARDS_NUMBER:1} # Shard number of new indexes
  indexReplicasNumber: ${SW_STORAGE_ES_INDEX_REPLICAS_NUMBER:1} # Replicas number of new indexes
  # Specify the settings for each index individually.
  # If configured, this setting has the highest priority and overrides the generic settings.
  specificIndexSettings: ${SW_STORAGE_ES_SPECIFIC_INDEX_SETTINGS:""}
  # Super data set has been defined in the codes, such as trace segments. The following 3 config would be improve es performance when storage super size data in es.
  superDataSetDayStep: ${SW_STORAGE_ES_SUPER_DATASET_DAY_STEP:1} # Represent the number of days in the super size dataset record index, the default value is the same as dayStep when the value is less than 0
  superDataSetIndexShardsFactor: ${SW_STORAGE_ES_SUPER_DATASET_INDEX_SHARDS_FACTOR:5} # This factor provides more shards for the super data set, shards number = indexShardsNumber * superDataSetIndexShardsFactor. Also,
  superDataSetIndexReplicasNumber: ${SW_STORAGE_ES_SUPER_DATASET_INDEX_REPLICAS_NUMBER:0} # Represent the replicas number in the super size dataset record index, the default value is 0.
  indexTemplateOrder: ${SW_STORAGE_ES_INDEX_TEMPLATE_ORDER:0} # the order of index template
  bulkActions: ${SW_STORAGE_ES_BULK_ACTIONS:15000} # Execute the async bulk record data every ${SW_STORAGE_ES_BULK_ACTIONS} requests
  # flush the bulk every 10 seconds whatever the number of requests
  # INT(flushInterval * 2/3) would be used for index refresh period.
  flushInterval: ${SW_STORAGE_ES_FLUSH_INTERVAL:15}
  concurrentRequests: ${SW_STORAGE_ES_CONCURRENT_REQUESTS:2} # the number of concurrent requests
  resultWindowMaxSize: ${SW_STORAGE_ES_QUERY_MAX_WINDOW_SIZE:10000}
  metaDataQueryMaxSize: ${SW_STORAGE_ES_QUERY_MAX_SIZE:10000}
  scrollingBatchSize: ${SW_STORAGE_ES_SCROLLING_BATCH_SIZE:5000}
  segmentQueryMaxSize: ${SW_STORAGE_ES_QUERY_SEGMENT_SIZE:200}
  profileTaskQueryMaxSize: ${SW_STORAGE_ES_QUERY_PROFILE_TASK_SIZE:200}
  profileDataQueryBatchSize: ${SW_STORAGE_ES_QUERY_PROFILE_DATA_BATCH_SIZE:100}
```


- 修改 **elasticsearch** 访问信息。

```
storage:
  selector: ${SW_STORAGE:elasticsearch}
  elasticsearch:
    namespace: ${SW_NAMESPACE:"elasticsearch"}
    clusterNodes: ${SW_STORAGE_ES_CLUSTER_NODES:localhost:9200}
    protocol: ${SW_STORAGE_ES_HTTP_PROTOCOL:"http"}
    connectTimeout: ${SW_STORAGE_ES_CONNECT_TIMEOUT:3000}
    socketTimeout: ${SW_STORAGE_ES_SOCKET_TIMEOUT:30000}
    responseTimeout: ${SW_STORAGE_ES_RESPONSE_TIMEOUT:15000}
    numHttpClientThread: ${SW_STORAGE_ES_NUM_HTTP_CLIENT_THREAD:0}
    user: ${SW_ES_USER:""}
    password: ${SW_ES_PASSWORD:""}
    trustStorePath: ${SW_STORAGE_ES_SSL_JKS_PATH:""}
    trustStorePass: ${SW_STORAGE_ES_SSL_JKS_PASS:""}
    secretsManagementFile: ${SW_ES_SECRETS_MANAGEMENT_FILE:""} # Secrets management file in the properties
    dayStep: ${SW_STORAGE_DAY_STEP:1} # Represent the number of days in the one minute/hour/day index.
    indexShardsNumber: ${SW_STORAGE_ES_INDEX_SHARDS_NUMBER:1} # Shard number of new indexes
    indexReplicasNumber: ${SW_STORAGE_ES_INDEX_REPLICAS_NUMBER:1} # Replicas number of new indexes
    # Specify the settings for each index individually.
```

```
storage:
  selector: ${SW_STORAGE:elasticsearch}
  elasticsearch:
    namespace: ${SW_NAMESPACE:""}
    clusterNodes: ${SW_STORAGE_ES_CLUSTER_NODES:localhost:9200}
    protocol: ${SW_STORAGE_ES_HTTP_PROTOCOL:"http"}
    connectTimeout: ${SW_STORAGE_ES_CONNECT_TIMEOUT:3000}
    socketTimeout: ${SW_STORAGE_ES_SOCKET_TIMEOUT:30000}
    responseTimeout: ${SW_STORAGE_ES_RESPONSE_TIMEOUT:15000}
    numHttpClientThread: ${SW_STORAGE_ES_NUM_HTTP_CLIENT_THREAD:0}
    user: ${SW_ES_USER:""}
    password: ${SW_ES_PASSWORD:""}
    trustStorePath: ${SW_STORAGE_ES_SSL_JKS_PATH:""}
    trustStorePass: ${SW_STORAGE_ES_SSL_JKS_PASS:""}
    secretsManagementFile: ${SW_ES_SECRETS_MANAGEMENT_FILE:""} # Secrets management file in the properties
    dayStep: ${SW_STORAGE_DAY_STEP:1} # Represent the number of days in the one minute/hour/day index.
    indexShardsNumber: ${SW_STORAGE_ES_INDEX_SHARDS_NUMBER:1} # Shard number of new indexes
    indexReplicasNumber: ${SW_STORAGE_ES_INDEX_REPLICAS_NUMBER:1} # Replicas number of new indexes
    # Specify the settings for each index individually.
```



- 安装示例中 **elasticsearch** 未进行配置修改，均为默认配置。
- **storage:elasticsearch:namespace** 默认为 **elasticsearch**。

6. 替换“skywalking-webapp.jar”文件。

使用产品工具包中提供的“skywalking-webapp.jar”替换“apache-skywalking-apm-bin\webapp”目录下“skywalking-webapp.jar”文件。

7. 启动 oap service。

- 进入“apache-skywalking-apm-bin\bin”目录。
- 运行“oapService.bat”脚本。

注意：

若用户修改了“apache-skywalking-apm-bin\config”目录下“application.yml”文件的“core:default:restPort”，则需要相应修改“apache-skywalking-apm-bin\webapp”目录下的“application.yml”配置文件中的“oapServices”配置项。

- core:default:restPort

```

internalComHost: ${SW_CLUSTER_INTERNAL_COM_HOST:""}
internalComPort: ${SW_CLUSTER_INTERNAL_COM_PORT:-1}
nacos:
  serviceName: ${SW_SERVICE_NAME:"SkyWalking_OAP_Cluster"}
  hostPort: ${SW_CLUSTER_NACOS_HOST_PORT:localhost:8848}
  # Nacos Configuration namespace
  namespace: ${SW_CLUSTER_NACOS_NAMESPACE:"public"}
  # Nacos auth username
  username: ${SW_CLUSTER_NACOS_USERNAME:""}
  password: ${SW_CLUSTER_NACOS_PASSWORD:""}
  # Nacos auth accessKey
  accessKey: ${SW_CLUSTER_NACOS_ACCESSKEY:""}
  secretKey: ${SW_CLUSTER_NACOS_SECRETKEY:""}
  internalComHost: ${SW_CLUSTER_INTERNAL_COM_HOST:""}
  internalComPort: ${SW_CLUSTER_INTERNAL_COM_PORT:-1}
core:
  selector: ${SW_CORE:default}
  default:
    # Mixed: Receive agent data, Level 1 aggregate, Level 2 aggregate
    # Receiver: Receive agent data, Level 1 aggregate
    # Aggregator: Level 2 aggregate
    role: ${SW_CORE_ROLE:Mixed} # Mixed/Receiver/Aggregator
    restHost: ${SW_CORE_REST_HOST:0.0.0.0}
    restPort: ${SW_CORE_REST_PORT:12801}
    restContextPath: ${SW_CORE_REST_CONTEXT_PATH:/}
    restMaxThreads: ${SW_CORE_REST_MAX_THREADS:200}
    restIdleTimeout: ${SW_CORE_REST_IDLE_TIMEOUT:30000}
    restAcceptQueueSize: ${SW_CORE_REST_QUEUE_SIZE:0}
    httpMaxRequestHeaderSize: ${SW_CORE_HTTP_MAX_REQUEST_HEADER_SIZE:8192}
    grpcHost: ${SW_CORE_GRPC_HOST:0.0.0.0}
    grpcPort: ${SW_CORE_GRPC_PORT:11801}
    maxConcurrentCallsPerConnection: ${SW_CORE_GRPC_MAX_CONCURRENT_CALL:0}

```

▪ oapServices

```

[esi@master01 webapp]$ cat application.yml
# Licensed to the Apache Software Foundation (ASF) under one or more
# contributor license agreements. See the NOTICE file distributed with
# this work for additional information regarding copyright ownership.
# The ASF licenses this file to You under the Apache License, Version 2.0
# (the "License"); you may not use this file except in compliance with
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# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

serverPort: ${SW_SERVER_PORT:-18181}

# Comma seperated list of OAP addresses.
oapServices: ${SW_OAP_ADDRESS:-http://localhost:12801}

```

8. 检查执行日志是否启动成功。

“apache-skywalking-apm-bin\logs” 目录下 “skywalking-oap-server.log”。

```
2023-09-04 10:33:18,896 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler ManagementServiceGrpcHandlerCompat into gRPC server 0.0.0.0:11800
2023-09-04 10:33:18,896 - org.apache.skywalking.oap.server.library.server.http.HTTPServer - 86 [main] INFO [] - Bind handler ManagementServiceHTTPHandler into http server 0.0.0.0:12800
2023-09-04 10:33:18,896 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in receiver-browser module.
2023-09-04 10:33:18,980 - org.apache.skywalking.oap.server.core.storage.model.ModelInstaller - 54 [main] INFO [] - table: browser_app_error_rate does not exist
2023-09-04 10:33:19,066 - org.apache.skywalking.oap.server.storage.plugin.elasticsearch.base.StorageEsInstaller - 188 [main] INFO [] - create metrics-all index template finished, isAcknowledged: true
2023-09-04 10:33:19,132 - org.apache.skywalking.oap.server.storage.plugin.elasticsearch.base.StorageEsInstaller - 202 [main] INFO [] - update metrics-all-20230904 index finished, isAcknowledged: true,
append mappings: Mappings={type=type, properties={numerator={type=long}, denominator={type=long}}, source=null)
2023-09-04 10:33:19,272 - org.apache.skywalking.oap.server.core.analysis.DispatcherManager - 117 [main] INFO [] - Dispatcher
org.apache.skywalking.oap.server.core.source.oal.rt.dispatcher.BrowserAppPagePerfDispatcher is added into DefaultScopeDefine 36.
2023-09-04 10:33:19,272 - org.apache.skywalking.oap.server.core.analysis.DispatcherManager - 117 [main] INFO [] - Dispatcher
org.apache.skywalking.oap.server.core.source.oal.rt.dispatcher.BrowserAppPageTrafficDispatcher is added into DefaultScopeDefine 40.
2023-09-04 10:33:19,272 - org.apache.skywalking.oap.server.core.analysis.DispatcherManager - 117 [main] INFO [] - Dispatcher
org.apache.skywalking.oap.server.core.source.oal.rt.dispatcher.BrowserAppSingleVersionTrafficDispatcher is added into DefaultScopeDefine 39.
2023-09-04 10:33:19,272 - org.apache.skywalking.oap.server.core.analysis.DispatcherManager - 117 [main] INFO [] - Dispatcher
org.apache.skywalking.oap.server.core.source.oal.rt.dispatcher.BrowserAppTrafficDispatcher is added into DefaultScopeDefine 38.
2023-09-04 10:33:19,274 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler BrowserPerfServiceHandler into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,276 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler BrowserPerfServiceHandlerCompat into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,277 - org.apache.skywalking.oap.server.library.server.http.HTTPServer - 86 [main] INFO [] - Bind handler BrowserPerfServiceHTTPHandler into http server 0.0.0.0:12800
2023-09-04 10:33:19,277 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in receiver-event module.
2023-09-04 10:33:19,277 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler EventGrpcServiceHandler into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,279 - org.apache.skywalking.oap.server.library.server.http.HTTPServer - 86 [main] INFO [] - Bind handler EventRestServiceHandler into http server 0.0.0.0:12800
2023-09-04 10:33:19,279 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in receiver-log module.
2023-09-04 10:33:19,279 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler LogReportServiceGrpcHandler into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,280 - org.apache.skywalking.oap.server.library.server.http.HTTPServer - 86 [main] INFO [] - Bind handler LogReportServiceHTTPHandler into http server 0.0.0.0:12800
2023-09-04 10:33:19,280 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in receiver-ebpf module.
2023-09-04 10:33:19,280 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler EBPFProcessServiceHandler into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,282 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler EBPFProfilingServiceHandler into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,284 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in receiver-meter module.
2023-09-04 10:33:19,284 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler MeterServiceHandler into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,285 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler MeterServiceHandlerCompat into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,286 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in envoy-metric module.
2023-09-04 10:33:19,900 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler MetricServiceGrpcHandler into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,904 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler MetricServiceGrpcHandlerV3 into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,905 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler AccessLogServiceGrpcHandler into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,909 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler AccessLogServiceGrpcHandlerV3 into gRPC server 0.0.0.0:11800
2023-09-04 10:33:19,910 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler SatelliteAccessLogServiceGrpcHandlerV3 into gRPC server 0.0.0.0:11800
2023-09-04 10:33:20,348 - com.linecorp.armeria.server.Server - 797 [armeria-boss-http-*:12800] INFO [] - Serving HTTP at /0:0:0:0:0:0:0:0:12800 - http://127.0.0.1:12800/
2023-09-04 10:33:20,349 - org.apache.skywalking.oap.server.core.cache.CacheUpdateTimer - 58 [main] INFO [] - persistence timer start
2023-09-04 10:33:20,350 - org.apache.skywalking.oap.server.core.cache.CacheUpdateTimer - 46 [main] INFO [] - Cache updateServiceInventory timer start
2023-09-04 10:33:25,306 - org.apache.skywalking.oap.server.starter.OAPServerBootstrap - 53 [main] INFO [] - Version of OAP: 9.3.0-7ad0406
```

9. 启动 webapp service。

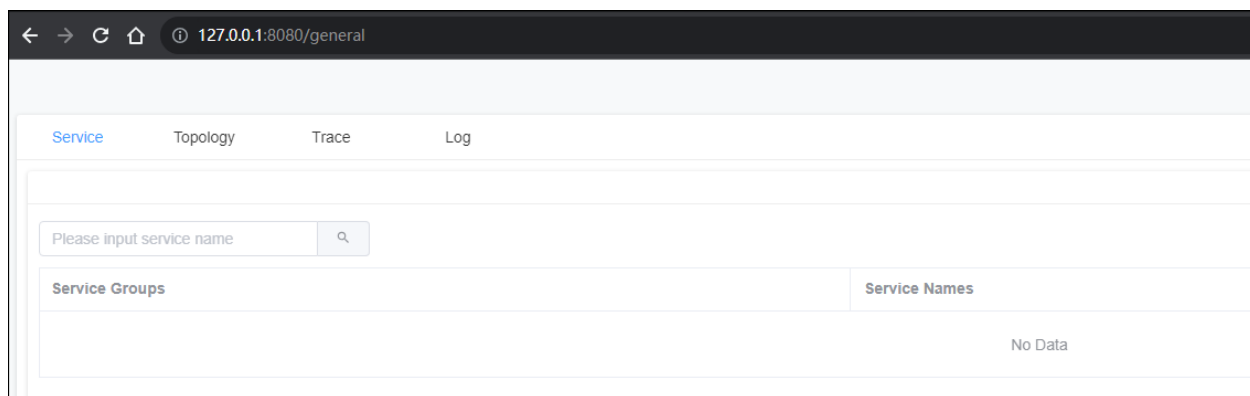
执行“apache-skywalking-apm-bin\bin”目录下“webappService.bat”。

10. 检查执行日志是否启动成功。

- 打开“apache-skywalking-apm-bin\logs”目录下“skywalking-webapp.log”。
- 获取访问地址。

```
2023-09-04 10:38:03,163 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxServerNumRequestsPerConnection: 0 (default)
2023-09-04 10:38:03,163 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxClientNumRequestsPerConnection: 0 (default)
2023-09-04 10:38:03,163 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxServerConnectionAgeMillis: 0 (default)
2023-09-04 10:38:03,164 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxClientConnectionAgeMillis: 0 (default)
2023-09-04 10:38:03,164 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultServerConnectionDrainDurationMicros: 1000000 (default)
2023-09-04 10:38:03,164 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp2InitialConnectionWindowSize: 1048576 (default)
2023-09-04 10:38:03,165 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp2InitialStreamWindowSize: 1048576 (default)
2023-09-04 10:38:03,165 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp2MaxFrameSize: 16384 (default)
2023-09-04 10:38:03,166 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp2MaxStreamsPerConnection: 2147483647 (default)
2023-09-04 10:38:03,166 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp2MaxHeaderListSize: 8192 (default)
2023-09-04 10:38:03,166 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp1MaxInitialLineLength: 4096 (default)
2023-09-04 10:38:03,167 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp1MaxHeaderSize: 8192 (default)
2023-09-04 10:38:03,167 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp1MaxChunkSize: 8192 (default)
2023-09-04 10:38:03,167 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultUseHttp2Preface: true (default)
2023-09-04 10:38:03,168 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultUseHttp1Pipelining: false (default)
2023-09-04 10:38:03,173 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultBackoffSpec: exponential=200:10000,jitter=0.2 (default)
2023-09-04 10:38:03,173 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxTotalAttempts: 10 (default)
2023-09-04 10:38:03,178 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - routeCacheSpec: maximumSize=4096 (default)
2023-09-04 10:38:03,178 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - routeDecoratorCacheSpec: maximumSize=4096 (default)
2023-09-04 10:38:03,178 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - parsedPathCacheSpec: maximumSize=4096 (default)
2023-09-04 10:38:03,179 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - headerValueCacheSpec: maximumSize=4096 (default)
2023-09-04 10:38:03,180 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - cachedHeaders: [authority, :scheme, :method, accept-encoding, content-type] (default)
2023-09-04 10:38:03,180 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - fileServiceCacheSpec: maximumSize=1024 (default)
2023-09-04 10:38:03,180 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - dnsCacheSpec: maximumSize=4096 (default)
2023-09-04 10:38:03,181 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - annotatedServiceExceptionVerbosity: unhandled (default)
2023-09-04 10:38:03,181 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - useJdkDnsResolver: false (default)
2023-09-04 10:38:03,181 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - reportBlockedEventLoop: true (default)
2023-09-04 10:38:03,182 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - reportMaskedRoutes: true (default)
2023-09-04 10:38:03,182 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - validateHeaders: true (default)
2023-09-04 10:38:03,182 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - tlsAllowUnsafeCiphers: false (default)
2023-09-04 10:38:03,182 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - transientServiceOptions: [] (default)
2023-09-04 10:38:03,183 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - useLegacyRouteDecoratorOrdering: false (default)
2023-09-04 10:38:03,183 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - useDefaultSocketOptions: true (default)
2023-09-04 10:38:03,183 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - allowDoubleDotsInQueryString: false (default)
2023-09-04 10:38:03,184 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMultipartUploadsLocation: C:\Users\dell\AppData\Local\Temp\armeria\multipart-uploads (default)
2023-09-04 10:38:03,886 - com.linecorp.armeria.common.util.SystemInfo - 525 [main] INFO [] - IPv6: disabled (no IPv6 network interface)
2023-09-04 10:38:03,929 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - useOpenSSL: true (default)
2023-09-04 10:38:04,223 - com.linecorp.armeria.common.Flags - 534 [main] INFO [] - Using OpenSSL: BoringSSL, 0x1010107f
2023-09-04 10:38:04,224 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - dumpOpenSSLInfo: false (default)
2023-09-04 10:38:04,496 - com.linecorp.armeria.common.util.SystemInfo - 260 [main] INFO [] - hostname: desktop-vj85cj6 (from 'hostname' command)
2023-09-04 10:38:04,541 - com.linecorp.armeria.server.Server - 797 [armeria-boss-http-*:8080] INFO [] - Serving HTTP at /0:0:0:0:0:0:0:0:8080 - http://127.0.0.1:8080/
```

- 在浏览器中，访问“http://IP:8080”。



1.1.3. 安装并启动 skywalking agent

本章节介绍如何安装并启动 skywalking agent。

前置条件

已获取产品工具包中的“apm-is-2.x-plugin-8.16.0.jar”。

“apm-is-2.x-plugin-8.16.0.jar”为产品提供的自定义插件。

操作步骤

1. 下载 [skywalking java agent 8.16.0](#)。
2. 解压下载的“apache-skywalking-java-agent-8.16.0.gz”。
3. 进入解压目录“skywalking java agent”。
4. 添加相关插件。

- **apm-spring-cloud-gateway-2.1.x-plugin-8.16.0.jar**

将解压目录中的“/optional-plugins/apm-spring-cloud-gateway-2.1.x-plugin-8.16.0.jar”复制到解压目录的“/plugins/”下。

- **apm-spring-webflux-5.x-plugin-8.16.0.jar**

将解压目录中的“/optional-plugins/apm-spring-webflux-5.x-plugin-8.16.0.jar”复制到解压目录的“/plugins/”下。

- **apm-is-2.x-plugin-8.16.0.jar**

注意：

- “apm-is-2.x-plugin-8.16.0.jar”为自定义插件，需要手动上传至“plugins”目录。
- 若用户修改了“apache-skywalking-apm-bin\config”目录下“application.yml”的“core:default:gRPCPort”，则需要相应修改环境变量“IS_SKYWALKING_SERVICE”。

core:default:gRPCPort

```

username: ${SW_CLUSTER_NACOS_USERNAME:""}
password: ${SW_CLUSTER_NACOS_PASSWORD:""}
# Nacos auth accessKey
accessKey: ${SW_CLUSTER_NACOS_ACCESSKEY:""}
secretKey: ${SW_CLUSTER_NACOS_SECRETKEY:""}
internalComHost: ${SW_CLUSTER_INTERNAL_COM_HOST:""}
internalComPort: ${SW_CLUSTER_INTERNAL_COM_PORT:-1}
core:
  selector: ${SW_CORE:default}
  default:
    # Mixed: Receive agent data, Level 1 aggregate, Level 2 aggregate
    # Receiver: Receive agent data, Level 1 aggregate
    # Aggregator: Level 2 aggregate
    role: ${SW_CORE_ROLE:Mixed} # Mixed/Receiver/Aggregator
    restHost: ${SW_CORE_REST_HOST:0.0.0.0}
    restPort: ${SW_CORE_REST_PORT:12801}
    restContextPath: ${SW_CORE_REST_CONTEXT_PATH:/}
    restMaxThreads: ${SW_CORE_REST_MAX_THREADS:200}
    restIdleTimeout: ${SW_CORE_REST_IDLE_TIMEOUT:30000}
    restAcceptQueueSize: ${SW_CORE_REST_QUEUE_SIZE:0}
    httpMaxRequestHeaderSize: ${SW_CORE_HTTP_MAX_REQUEST_HEADER_SIZE:8192}
    grpcHost: ${SW_CORE_GRPC_HOST:0.0.0.0}
    grpcPort: ${SW_CORE_GRPC_PORT:11801}
    maxConcurrentCallsPerConnection: ${SW_CORE_GRPC_MAX_CONCURRENT_CALL:0}
    maxMessageSize: ${SW_CORE_GRPC_MAX_MESSAGE_SIZE:0}
    grpcThreadPoolQueueSize: ${SW_CORE_GRPC_POOL_QUEUE_SIZE:-1}
    grpcThreadPoolSize: ${SW_CORE_GRPC_THREAD_POOL_SIZE:-1}
    grpcSslEnabled: ${SW_CORE_GRPC_SSL_ENABLED:false}
    grpcSslKeyPath: ${SW_CORE_GRPC_SSL_KEY_PATH:""}
    grpcSslCertChainPath: ${SW_CORE_GRPC_SSL_CERT_CHAIN_PATH:""}
    grpcSslTrustedCAPath: ${SW_CORE_GRPC_SSL_TRUSTED_CA_PATH:""}
    downsampling:

```

环境变量 IS_SKYWALKING_SERVICE

```

[esi@master01 apache-skywalking-apm-bin]$ env | grep IS
IS_SKYWALKING_AGENT_HOME=/home/esi/skywalking/skywalking-agent/
IS_AM_PORT=18097
IS_PROXY_HTTP_PORT=8185
IS_SERVER_IP_ADDR=10.10.81.35
IS_AM_IP=10.10.81.35
IS_SKYWALKING_WEBSITE_URL=http://10.10.81.35:18180
IS_MANAGE_HTTP_PORT=9991
IS_DB_USER=root
IS_SERVER_HTTP_PORT=19612
IS_DB_TYPE=mysql
IS_DB_SCHEMA=
IS_DB_PASSWORD=123456
IS_REDIS_IP=10.10.81.35
IS_SKYWALKING_ENABLE=true
IS_DB_URL=jdbc:mysql://10.10.81.35:3306/trongis?allowMultiQueries=true&useSSL=false&serverTimezone=Asia/Shanghai&rewriteBatchedStatements=true&characterEncoding=utf8
IS_SKYWALKING_SERVICE=127.0.0.1:11800
IS_REDIS_PASSWORD=
IS_REDIS_PORT=26379
IS_REDIS_DB=15
IS_DB_DRIVER=com.mysql.cj.jdbc.Driver

```

1.2. Linux平台

1.2.1. 安装并启动 Elasticsearch

安装 Skywalking 之前，需要先按照 ElasticSearch 作存储。

如下以“Elasticsearch Linux 7.8.0”为例说明。

1. 下载 [Elasticsearch Linux 7.8.0](#)。
2. 解压下载的 Elasticsearch 安装包。
3. 进入解压目录“elasticsearch-7.8.0/bin”。
4. 执行如下命令，启动 Elasticsearch。

```
./elasticsearch-7.8.0/bin/elasticsearch
```

5. 在浏览器中，访问如下地址，检查 Elasticsearch 是否启动成功。

```
http://IP:9200/
```

说明：

IP: 表示安装 Elasticsearch 的主机的 IP 地址。

```
[esi@master01 ~]$ curl 127.0.0.1:9201
{
  "name" : "master01",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "Jzi3KMxdS7GZhq8WlxuDjQ",
  "version" : {
    "number" : "7.8.0",
    "build_flavor" : "default",
    "build_type" : "tar",
    "build_hash" : "757314695644ea9a1dc2fec26d1a43856725e65",
    "build_date" : "2020-06-14T19:35:50.234439Z",
    "build_snapshot" : false,
    "lucene_version" : "8.5.1",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
```

1.2.2. 安装并启动 skywalking

本章节介绍如何安装并启动 skywalking。

注意事项

Skywalking 安装目录命名不要使用空格，使用空格容易导致启动失败。

前置条件

已获取产品工具包中的“skywalking-webapp.jar”。

“skywalking-webapp.jar”针对管理中心进行了部分定制化。

操作步骤

1. 下载 [skywalking apm 9.3.0](#)。
2. 解压下载的 skywalking 安装包。
3. 进入“apache-skywalking-apm-bin\config”目录。
4. 打开“application.yml”文件。
5. 修改如下配置信息。
 - 将“storage:selector”修改为“elasticsearch”。

```
storage:
  selector: ${SW_STORAGE:elasticsearch}
  elasticsearch:
    namespace: ${SW_NAMESPACE:""}
    clusterNodes: ${SW_STORAGE_ES_CLUSTER_NODES:localhost:9201}
    protocol: ${SW_STORAGE_ES_HTTP_PROTOCOL:"http"}
    connectTimeout: ${SW_STORAGE_ES_CONNECT_TIMEOUT:3000}
    socketTimeout: ${SW_STORAGE_ES_SOCKET_TIMEOUT:30000}
    responseTimeout: ${SW_STORAGE_ES_RESPONSE_TIMEOUT:15000}
    numHttpClientThread: ${SW_STORAGE_ES_NUM_HTTP_CLIENT_THREAD:0}
    user: ${SW_ES_USER:""}
    password: ${SW_ES_PASSWORD:""}
    trustStorePath: ${SW_STORAGE_ES_SSL_JKS_PATH:""}
    trustStorePass: ${SW_STORAGE_ES_SSL_JKS_PASS:""}
    secretsManagementFile: ${SW_ES_SECRETS_MANAGEMENT_FILE:""} # Secrets management file in the properties format
    d party tool.
    dayStep: ${SW_STORAGE_DAY_STEP:1} # Represent the number of days in the one minute/hour/day index.
    indexShardsNumber: ${SW_STORAGE_ES_INDEX_SHARDS_NUMBER:1} # Shard number of new indexes
    indexReplicasNumber: ${SW_STORAGE_ES_INDEX_REPLICAS_NUMBER:1} # Replicas number of new indexes
```

- 修改 elasticsearch 访问信息。

```

prepareThreads: ${SW_CORE_PREPARE_THREADS:2}
# Turn it on then automatically grouping endpoint by the given OpenAPI definitions.
enableEndpointNameGroupingByOpenapi: ${SW_CORE_ENABLE_ENDPOINT_NAME_GROUPING_BY_OPENAPI:true}
storage:
  selector: ${SW_STORAGE:elasticsearch}
  elasticsearch:
    namespace: ${SW_NAMESPACE:"elasticsearch"}
    clusterNodes: ${SW_STORAGE_ES_CLUSTER_NODES:localhost:9201}
    protocol: ${SW_STORAGE_ES_HTTP_PROTOCOL:http}
    connectTimeout: ${SW_STORAGE_ES_CONNECT_TIMEOUT:3000}
    socketTimeout: ${SW_STORAGE_ES_SOCKET_TIMEOUT:30000}
    responseTimeout: ${SW_STORAGE_ES_RESPONSE_TIMEOUT:15000}
    numHttpClientThread: ${SW_STORAGE_ES_NUM_HTTP_CLIENT_THREAD:0}
    user: ${SW_ES_USER:""}
    password: ${SW_ES_PASSWORD:""}
    trustStorePath: ${SW_STORAGE_ES_SSL_JKS_PATH:""}
    trustStorePass: ${SW_STORAGE_ES_SSL_JKS_PASS:""}
    secretsManagementFile: ${SW_ES_SECRETS_MANAGEMENT_FILE:""} # Secrets management file in the properties
    d party tool.

autocompleteTagValuesQueryMaxSize: ${SW_AUTOCOMPLETE_TAG_VALUES_QUERY_MAX_SIZE:100}
# The number of threads used to prepare metrics data to the storage.
prepareThreads: ${SW_CORE_PREPARE_THREADS:2}
# Turn it on then automatically grouping endpoint by the given OpenAPI definitions.
enableEndpointNameGroupingByOpenapi: ${SW_CORE_ENABLE_ENDPOINT_NAME_GROUPING_BY_OPENAPI:true}
storage:
  selector: ${SW_STORAGE:elasticsearch}
  elasticsearch:
    namespace: ${SW_NAMESPACE:"elasticsearch"}
    clusterNodes: ${SW_STORAGE_ES_CLUSTER_NODES:localhost:9201}
    protocol: ${SW_STORAGE_ES_HTTP_PROTOCOL:http}
    connectTimeout: ${SW_STORAGE_ES_CONNECT_TIMEOUT:3000}
    socketTimeout: ${SW_STORAGE_ES_SOCKET_TIMEOUT:30000}
    responseTimeout: ${SW_STORAGE_ES_RESPONSE_TIMEOUT:15000}
    numHttpClientThread: ${SW_STORAGE_ES_NUM_HTTP_CLIENT_THREAD:0}
    user: ${SW_ES_USER:""}
    password: ${SW_ES_PASSWORD:""}
    trustStorePath: ${SW_STORAGE_ES_SSL_JKS_PATH:""}
    trustStorePass: ${SW_STORAGE_ES_SSL_JKS_PASS:""}
    secretsManagementFile: ${SW_ES_SECRETS_MANAGEMENT_FILE:""} # Secrets management file in the properties
    d party tool.

```



- 安装示例中 **elasticsearch** 未进行配置修改，均为默认配置。
- **storage:elasticsearch:namespace** 默认为 **elasticsearch**。

6. 替换“skywalking-webapp.jar”文件。

使用产品工具包中提供的“skywalking-webapp.jar”替换“apache-skywalking-apm-bin\webapp”目录下“skywalking-webapp.jar”文件。

7. 启动 oap service。

- 进入“apache-skywalking-apm-bin\bin”目录。
- 运行“oapService.sh”脚本，如下图所示。

```

[esi@master01 bin]$ ./oapService.sh
SkyWalking OAP started successfully!
[esi@master01 bin]$

```

8. 检查执行日志是否启动成功。

“apache-skywalking-apm-bin\logs”目录下“skywalking-oap-server.log”。

```

RPC server 0.0.0.0:11801
2023-09-04 14:06:36,709 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in receiver-register module.
2023-09-04 14:06:36,709 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler ManagementServiceGRPCHandler into gRPC server 0.0.0.0:11801
2023-09-04 14:06:36,713 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler ManagementServiceGrpcHandlerCompat into gRPC server 0.0.0.0:11801
2023-09-04 14:06:36,715 - org.apache.skywalking.oap.server.library.server.http.HTTPServer - 86 [main] INFO [] - Bind handler ManagementServiceHTTPHandler into http server 0.0.0.0:12801
2023-09-04 14:06:36,715 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in receiver-browser module.
2023-09-04 14:06:37,003 - org.apache.skywalking.oap.server.core.analysis.DispatcherManager - 117 [main] INFO [] - Dispatcher org.apache.skywalking.oap.server.core.source.oal.rt.dispatcher.BrowserAppPagePerfDispatcher is added into DefaultScopeDefine 36.
2023-09-04 14:06:37,003 - org.apache.skywalking.oap.server.core.analysis.DispatcherManager - 117 [main] INFO [] - Dispatcher org.apache.skywalking.oap.server.core.source.oal.rt.dispatcher.BrowserAppPageTrafficDispatcher is added into DefaultScopeDefine 40.
2023-09-04 14:06:37,004 - org.apache.skywalking.oap.server.core.analysis.DispatcherManager - 117 [main] INFO [] - Dispatcher org.apache.skywalking.oap.server.core.source.oal.rt.dispatcher.BrowserAppSingleVersionTrafficDispatcher is added into DefaultScopeDefine 39.
2023-09-04 14:06:37,004 - org.apache.skywalking.oap.server.core.analysis.DispatcherManager - 117 [main] INFO [] - Dispatcher org.apache.skywalking.oap.server.core.source.oal.rt.dispatcher.BrowserAppTrafficDispatcher is added into DefaultScopeDefine 38.
2023-09-04 14:06:37,006 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler BrowserPerfServiceHandler into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,010 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler BrowserPerfServiceHandlerCompat into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,013 - org.apache.skywalking.oap.server.library.server.http.HTTPServer - 86 [main] INFO [] - Bind handler BrowserPerfServiceHTTPHandler into http server 0.0.0.0:12801
2023-09-04 14:06:37,013 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in receiver-event module.
2023-09-04 14:06:37,013 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler EventGrpcServiceHandler into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,017 - org.apache.skywalking.oap.server.library.server.http.HTTPServer - 86 [main] INFO [] - Bind handler EventRestServiceHandler into http server 0.0.0.0:12801
2023-09-04 14:06:37,017 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in receiver-log module.
2023-09-04 14:06:37,018 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler LogReportServiceGrpcHandler into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,020 - org.apache.skywalking.oap.server.library.server.http.HTTPServer - 86 [main] INFO [] - Bind handler LogReportServiceHTTPHandler into http server 0.0.0.0:12801
2023-09-04 14:06:37,020 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in receiver-ebpf module.
2023-09-04 14:06:37,020 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler EBPFProcessServiceHandler into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,025 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler EBPFProfilingServiceHandler into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,029 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in receiver-meter module.
2023-09-04 14:06:37,029 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler MetersServiceHandler into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,032 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler MetersServiceHandlerCompat into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,034 - org.apache.skywalking.oap.server.library.module.BootstrapFlow - 43 [main] INFO [] - start the provider default in envoy-metric module.
2023-09-04 14:06:37,659 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler MetricServiceGRPCHandler into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,663 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler MetricServiceGRPCHandlerV3 into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,664 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler AccessLogServiceGRPCHandler into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,668 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler AccessLogServiceGRPCHandlerV3 into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,669 - org.apache.skywalking.oap.server.library.server.grpc.GRPCServer - 131 [main] INFO [] - Bind handler SatelliteAccessLogServiceGRPCHandlerV3 into gRPC server 0.0.0.0:11801
2023-09-04 14:06:37,705 - com.linecorp.armeria.common.util.SystemInfo - 237 [main] INFO [] - hostname: master01 (from /proc/sys/kernel/hostname)
2023-09-04 14:06:38,312 - com.linecorp.armeria.server.Server - 797 [armeria-boss-http-*:12801] INFO [] - Serving HTTP at /0:0:0:0:0:0:0:0:12801 - http://127.0.0.1:12801/
2023-09-04 14:06:38,313 - org.apache.skywalking.oap.server.core.storage.PersistenceTimer - 58 [main] INFO [] - persistence timer start
2023-09-04 14:06:38,315 - org.apache.skywalking.oap.server.core.cache.CacheUpdateTimer - 46 [main] INFO [] - Cache updateServiceInventory timer start

```

9. 启动 webapp service。

执行“apache-skywalking-apm-bin\bin”目录下“webappService.sh”。

```

[esi@master01 apache-skywalking-apm-bin]$ ./bin/webappService.sh
skyWalking Web Application started successfully!
[esi@master01 apache-skywalking-apm-bin]$

```

注意：

若用户修改了“apache-skywalking-apm-bin\config”目录中“application.yml”文件的“core:default:restPort”，则需要相应修改“apache-skywalking-apm-bin\webapp”目录下的“application.yml”配置文件中的“oapServices”配置项。

- core:default:restPort


```

internalComHost: ${SW_CLUSTER_INTERNAL_COM_HOST:""}
internalComPort: ${SW_CLUSTER_INTERNAL_COM_PORT:-1}
nacos:
  serviceName: ${SW_SERVICE_NAME:"Skywalking_OAP_Cluster"}
  hostPort: ${SW_CLUSTER_NACOS_HOST_PORT:localhost:8848}
  # Nacos Configuration namespace
  namespace: ${SW_CLUSTER_NACOS_NAMESPACE:"public"}
  # Nacos auth username
  username: ${SW_CLUSTER_NACOS_USERNAME:""}
  password: ${SW_CLUSTER_NACOS_PASSWORD:""}
  # Nacos auth accessKey
  accessKey: ${SW_CLUSTER_NACOS_ACCESSKEY:""}
  secretKey: ${SW_CLUSTER_NACOS_SECRETKEY:""}
  internalComHost: ${SW_CLUSTER_INTERNAL_COM_HOST:""}
  internalComPort: ${SW_CLUSTER_INTERNAL_COM_PORT:-1}
core:
  selector: ${SW_CORE:default}
  default:
    # Mixed: Receive agent data, Level 1 aggregate, Level 2 aggregate
    # Receiver: Receive agent data, Level 1 aggregate
    # Aggregator: Level 2 aggregate
    role: ${SW_CORE_ROLE:Mixed} # Mixed/Receiver/Aggregator
    restHost: ${SW_CORE_REST_HOST:0.0.0.0}
    restPort: ${SW_CORE_REST_PORT:12801}
    restContextPath: ${SW_CORE_REST_CONTEXT_PATH:/}
    restMaxThreads: ${SW_CORE_REST_MAX_THREADS:200}
    restIdleTimeout: ${SW_CORE_REST_IDLE_TIMEOUT:30000}
    restAcceptQueueSize: ${SW_CORE_REST_QUEUE_SIZE:0}
    httpMaxRequestHeaderSize: ${SW_CORE_HTTP_MAX_REQUEST_HEADER_SIZE:8192}
    grpcHost: ${SW_CORE_GRPC_HOST:0.0.0.0}
    grpcPort: ${SW_CORE_GRPC_PORT:11801}
    maxConcurrentCallsPerConnection: ${SW_CORE_GRPC_MAX_CONCURRENT_CALL:0}

```

◦ oapServices

```

[esi@master01 webapp]$ cat application.yml
# Licensed to the Apache Software Foundation (ASF) under one or more
# contributor license agreements.  See the NOTICE file distributed with
# this work for additional information regarding copyright ownership.
# The ASF licenses this file to You under the Apache License, Version 2.0
# (the "License"); you may not use this file except in compliance with
# the License.  You may obtain a copy of the License at
#
#     http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

serverPort: ${SW_SERVER_PORT:-18181}

# Comma seperated list of OAP addresses.
oapServices: ${SW_OAP_ADDRESS:-http://localhost:12801}

```

10. 检查执行日志是否启动成功。

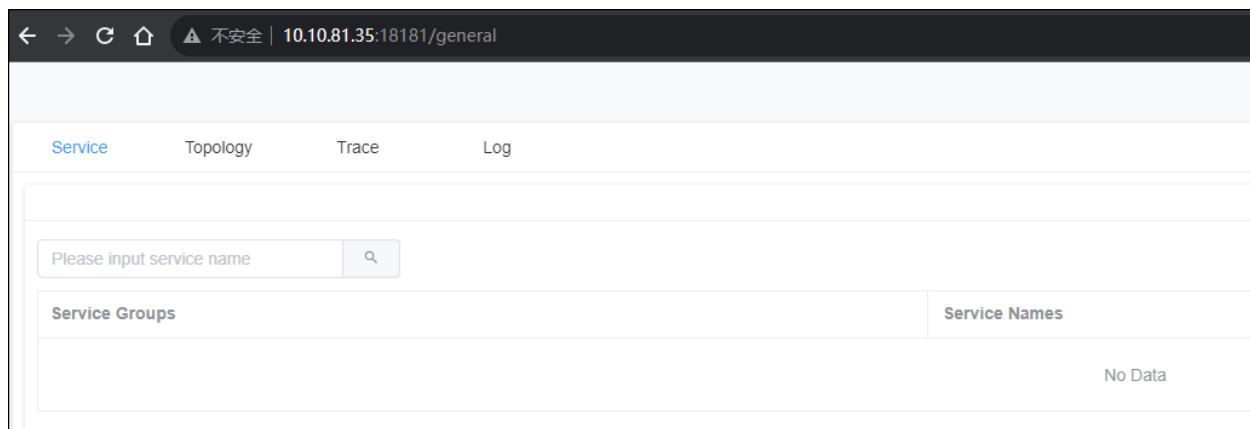
- a. 打开“apache-skywalking-apm-bin\logs”目录下“skywalking-webapp.log”。

```

2023-09-04 14:13:01.150 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - warnNettyVersions: true (default)
2023-09-04 14:13:01.243 - com.linecorp.armeria.common.Flags - 1373 [main] INFO [] - useEpoll: true (default)
2023-09-04 14:13:01.245 - com.linecorp.armeria.common.Flags - 174 [main] INFO [] - Using /dev/epoll
2023-09-04 14:13:01.245 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - transportType: EPOLL (default)
2023-09-04 14:13:01.246 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - maxNumConnections: 2147483647 (default)
2023-09-04 14:13:01.246 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - numCommonWorkers: 96 (default)
2023-09-04 14:13:01.247 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - numCommonBlockingTaskThreads: 200 (default)
2023-09-04 14:13:01.248 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxRequestLength: 10485760 (default)
2023-09-04 14:13:01.249 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxResponseLength: 10485760 (default)
2023-09-04 14:13:01.249 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultRequestTimeoutMillis: 10000 (default)
2023-09-04 14:13:01.250 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultResponseTimeoutMillis: 15000 (default)
2023-09-04 14:13:01.251 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultConnectTimeoutMillis: 3200 (default)
2023-09-04 14:13:01.251 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultWriteTimeoutMillis: 1000 (default)
2023-09-04 14:13:01.252 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultServerIdleTimeoutMillis: 15000 (default)
2023-09-04 14:13:01.253 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultClientIdleTimeoutMillis: 10000 (default)
2023-09-04 14:13:01.254 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultPingIntervalMillis: 0 (default)
2023-09-04 14:13:01.254 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxServerNumRequestsPerConnection: 0 (default)
2023-09-04 14:13:01.255 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxClientNumRequestsPerConnection: 0 (default)
2023-09-04 14:13:01.256 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxServerConnectionAgeMillis: 0 (default)
2023-09-04 14:13:01.256 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxClientConnectionAgeMillis: 0 (default)
2023-09-04 14:13:01.257 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultServerConnectionDrainDurationMicros: 1000000 (default)
2023-09-04 14:13:01.258 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp2InitialWindowSize: 1048576 (default)
2023-09-04 14:13:01.258 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp2InitialStreamWindowSize: 1048576 (default)
2023-09-04 14:13:01.259 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp2MaxFrameSize: 16384 (default)
2023-09-04 14:13:01.259 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp2MaxStreamsPerConnection: 2147483647 (default)
2023-09-04 14:13:01.260 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp2MaxHeaderListSize: 8192 (default)
2023-09-04 14:13:01.261 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp1MaxInitialLineLength: 4096 (default)
2023-09-04 14:13:01.261 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp1MaxHeaderSize: 8192 (default)
2023-09-04 14:13:01.262 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultHttp1MaxChunkSize: 8192 (default)
2023-09-04 14:13:01.262 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultUseHttp2Preface: true (default)
2023-09-04 14:13:01.263 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultUseHttp1Pipelining: false (default)
2023-09-04 14:13:01.271 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultBackoffSpec: exponential=200:10000,jitter=0.2 (default)
2023-09-04 14:13:01.272 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMaxTotalAttempts: 10 (default)
2023-09-04 14:13:01.281 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - routeCacheSpec: maximumSize=4096 (default)
2023-09-04 14:13:01.281 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - routeDecoratorCacheSpec: maximumSize=4096 (default)
2023-09-04 14:13:01.282 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - parsedPathCacheSpec: maximumSize=4096 (default)
2023-09-04 14:13:01.282 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - headerValueCacheSpec: maximumSize=4096 (default)
2023-09-04 14:13:01.284 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - cachedHeaders: [authority, :scheme, :method, accept-encoding, content-type] (default)
2023-09-04 14:13:01.285 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - fileServiceCacheSpec: maximumSize=1024 (default)
2023-09-04 14:13:01.285 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - dnsCacheSpec: maximumSize=4096 (default)
2023-09-04 14:13:01.285 - com.linecorp.armeria.common.Flags - 1373 [main] INFO [] - annotatedServiceExceptionVerbosity: unhandled (default)
2023-09-04 14:13:01.286 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - useJdkDnsResolver: false (default)
2023-09-04 14:13:01.287 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - reportLockedEventLoop: true (default)
2023-09-04 14:13:01.287 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - reportMaskedRoutes: true (default)
2023-09-04 14:13:01.287 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - validateHeaders: true (default)
2023-09-04 14:13:01.288 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - tlsAllowUnsafeCiphers: false (default)
2023-09-04 14:13:01.288 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - transientServiceOptions: [] (default)
2023-09-04 14:13:01.289 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - useLegacyRouteDecoratorOrdering: false (default)
2023-09-04 14:13:01.289 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - useDefaultSocketOptions: true (default)
2023-09-04 14:13:01.290 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - allowDoubleQuotesInQueryString: false (default)
2023-09-04 14:13:01.290 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - defaultMultipartUploadsLocation: /tmp/armeria/multipart-uploads (default)
2023-09-04 14:13:01.661 - com.linecorp.armeria.common.util.SystemInfo - 525 [main] INFO [] - IPv6: disabled (no IPv6 network interface)
2023-09-04 14:13:01.747 - com.linecorp.armeria.common.Flags - 1393 [main] INFO [] - useOpenSsl: true (default)
2023-09-04 14:13:01.943 - com.linecorp.armeria.common.Flags - 534 [main] INFO [] - Using OpenSSL: BoringSSL, 0x1010107f

```

b. 在浏览器中，访问“http://ip:8080”。



1.2.3. 安装并启动 skywalking agent

本章节介绍如何安装并启动 skywalking agent。

前置条件

已获取产品工具包中的“apm-is-2.x-plugin-8.16.0.jar”。

“apm-is-2.x-plugin-8.16.0.jar”为产品提供的自定义插件。

操作步骤

1. 下载 [skywalking java agent 8.16.0](#)。
2. 解压下载的“apache-skywalking-java-agent-8.16.0.gz”。

```
tar -zxvf apache-skywalking-java-agent-8.16.0.gz
```

3. 进入解压目录“skywalking java agent”。

```
cd skywalking-agent
```

4. 添加相关插件。

◦ **apm-spring-cloud-gateway-2.1.x-plugin-8.16.0.jar**

将解压目录中的“/optional-plugins/apm-spring-cloud-gateway-2.1.x-plugin-8.16.0.jar”复制到解压目录的“/plugins/”下。

```
cp ./optional-plugins/apm-spring-cloud-gateway-2.1.x-plugin-8.16.0.jar ./plugins/apm-spring-cloud-gateway-2.1.x-plugin-8.16.0.jar
```

◦ **apm-spring-webflux-5.x-plugin-8.16.0.jar**

将解压目录中的“/optional-plugins/apm-spring-webflux-5.x-plugin-8.16.0.jar”复制到解压目录的“/plugins/”下。

```
cp ./optional-plugins/apm-spring-webflux-5.x-plugin-8.16.0.jar ./plugins/apm-spring-webflux-5.x-plugin-8.16.0.jar
```

◦ **apm-is-2.x-plugin-8.16.0.jar**

注意：

- “apm-is-2.x-plugin-8.16.0.jar”为自定义插件，需要手动上传至“plugins”目录。
- 若用户修改了“apache-skywalking-apm-bin\config”目录中“application.yml”文件中的“core:default:gRPCPort”，则需要相应修改环境变量“IS_SKYWALKING_SERVICE”。

core:default:gRPCPort

```
username: ${SW_CLUSTER_NACOS_USERNAME:""}
password: ${SW_CLUSTER_NACOS_PASSWORD:""}
# Nacos auth accessKey
accessKey: ${SW_CLUSTER_NACOS_ACCESSKEY:""}
secretKey: ${SW_CLUSTER_NACOS_SECRETKEY:""}
internalComHost: ${SW_CLUSTER_INTERNAL_COM_HOST:""}
internalComPort: ${SW_CLUSTER_INTERNAL_COM_PORT:-1}
core:
  selector: ${SW_CORE:default}
  default:
    # Mixed: Receive agent data, Level 1 aggregate, Level 2 aggregate
    # Receiver: Receive agent data, Level 1 aggregate
    # Aggregator: Level 2 aggregate
    role: ${SW_CORE_ROLE:Mixed} # Mixed/Receiver/Aggregator
    restHost: ${SW_CORE_REST_HOST:0.0.0.0}
    restPort: ${SW_CORE_REST_PORT:12801}
    restContextPath: ${SW_CORE_REST_CONTEXT_PATH:/}
    restMaxThreads: ${SW_CORE_REST_MAX_THREADS:200}
    restIdleTimeout: ${SW_CORE_REST_IDLE_TIMEOUT:30000}
    restAcceptQueueSize: ${SW_CORE_REST_QUEUE_SIZE:0}
    httpMaxRequestHeaderSize: ${SW_CORE_HTTP_MAX_REQUEST_HEADER_SIZE:8192}
    gRPCHost: ${SW_CORE_GRPC_HOST:0.0.0.0}
    gRPCPort: ${SW_CORE_GRPC_PORT:11801}
    maxConcurrentCallsPerConnection: ${SW_CORE_GRPC_MAX_CONCURRENT_CALL:0}
    maxMessageSize: ${SW_CORE_GRPC_MAX_MESSAGE_SIZE:0}
    gRPCThreadPoolQueueSize: ${SW_CORE_GRPC_POOL_QUEUE_SIZE:-1}
    gRPCThreadPoolSize: ${SW_CORE_GRPC_THREAD_POOL_SIZE:-1}
    gRPCSslEnabled: ${SW_CORE_GRPC_SSL_ENABLED:false}
    gRPCSslKeyPath: ${SW_CORE_GRPC_SSL_KEY_PATH:""}
    gRPCSslCertChainPath: ${SW_CORE_GRPC_SSL_CERT_CHAIN_PATH:""}
    gRPCSslTrustedCAPath: ${SW_CORE_GRPC_SSL_TRUSTED_CA_PATH:""}
    downsampling:
```

环境变量 IS_SKYWALKING_SERVICE

```

esi@master01 apache-skywalking-apm-bin$ env | grep IS
IS_SKYWALKING_AGENT_HOME=/home/esi/skywalking/skywalking-agent/
IS_AM_PORT=18097
IS_PROXY_HTTP_PORT=8185
IS_SERVER_IP_ADDR=10.10.81.35
IS_AM_IP=10.10.81.35
IS_SKYWALKING_WEBSITE_URL=http://10.10.81.35:18180
IS_MANAGE_HTTP_PORT=9991
IS_DB_USER=root
IS_SERVER_HTTP_PORT=19612
IS_DB_TYPE=mysql
IS_DB_SCHEMA=
IS_DB_PASSWORD=123456
IS_REDIS_IP=10.10.81.35
IS_SKYWALKING_ENABLE=true
IS_DB_URL=jdbc:mysql://10.10.81.35:3306/trongis?allowMultiQueries=true&useSSL=false&serverTimezone=Asia/Shanghai&rewriteBatchedStatements=true&characterEncoding=utf8
IS_SKYWALKING_SERVICE=127.0.0.1:11800
IS_REDIS_PASSWORD=
IS_REDIS_PORT=26379
IS_REDIS_DB=15
IS_DB_DRIVER=com.mysql.cj.jdbc.Driver

```

1.3. Docker



Docker 均在 linux 环境下操作。

1.3.1. 安装并启动 Elasticsearch

安装 Skywalking 之前，需要先按照 ElasticSearch 作存储。

1. 执行如下命令，拉取 elasticsearch 7.8.0 镜像。

```
docker pull elasticsearch:7.8.0
```

2. 执行如下命令，安装并启动 elasticsearch。

```
docker run -d --name elasticsearch -p 9200:9200 -p 9300:9300 -e "discovery.type=single-node"
elasticsearch:7.8.0
```

3. 执行如下命令，检查是否启动成功。

```
curl http://ip:9200/
```

执行以上指令后，若返回如下信息，则说明启动成功。

```

{
  "name" : "f156442c579a",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "2Efi7xuTQNawm4cybnD7sg",
  "version" : {
    "number" : "7.8.0",
    "build_flavor" : "default",
    "build_type" : "docker",
    "build_hash" : "757314695644ea9a1dc2fec26d1a43856725e65",
    "build_date" : "2020-06-14T19:35:50.234439Z",
    "build_snapshot" : false,
    "lucene_version" : "8.5.1",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}

```

```
}
```

注意:

“9200”为 elasticsearch 默认访问端口。docker 启动 elasticsearch 时，可根据实际需要进行调整，以便后续启动 “skywalking oap server” 时使用。

1.3.2. 安装并启动 skywalking

本章节介绍如何安装并启动 skywalking。

注意事项

Skywalking 安装目录命名不要使用空格，使用空格容易导致启动失败。

前置条件

已获取产品工具包中的 “skywalking-webapp.jar”。

“skywalking-webapp.jar” 针对管理中心进行了部分定制化。

操作步骤

1. 执行如下命令，拉取 “skywalking oap server” 镜像。

```
docker pull apache/skywalking-oap-server:9.3.0
```

2. 执行如下命令，安装并启动 “skywalking oap server”。

```
docker run --name oap --restart always -d -p 11234:1234 -p 11800:11800 -p 12800:12800 -e  
SW_ENABLE_UPDATE_UI_TEMPLATE=true -e SW_STORAGE=elasticsearch -e  
SW_STORAGE_ES_CLUSTER_NODES=IP:9200 apache/skywalking-oap-server:9.3.0
```

注意:

- 将 “SW_STORAGE_ES_CLUSTER_NODES=IP:9200” 中 “IP” 替换为 “elasticsearch” 实际安装的地址。
 - “11800” 端口为 “skywalking agent” 上报信息时访问的端口，运行时及代理启动上报信息时将会使用。
 - “12800” 端口为 “skywalking ui” 访问端口，启动 “skywalking ui” 时将会使用。
 - “SW_STORAGE_ES_CLUSTER_NODES” 环境变量，指定 “skywalking” 将数据存储在 “elasticsearch” 中。
3. 执行如下命令，拉取 “skywalking ui” 镜像。

```
docker pull apache/skywalking-ui:9.3.0
```

4. 安装并启动 “skywalking ui”。

```
docker run --name oap_ui --restart always -d -p 18180:8080 -e SW_SERVER_PORT=8080 -e  
SW_OAP_ADDRESS=http://IP:12800 apache/skywalking-ui:9.3.0
```

注意:

将“SW_OAP_ADDRESS=http://IP:12800”中“IP”替换为“oap”实际安装的地址。

5. 替换 skywalking ui 的“skywalking-webapp.jar”文件。

```
docker cp ./skywalking-webapp.jar oap_ui:/skywalking/webapp/skywalking-webapp.jar
```

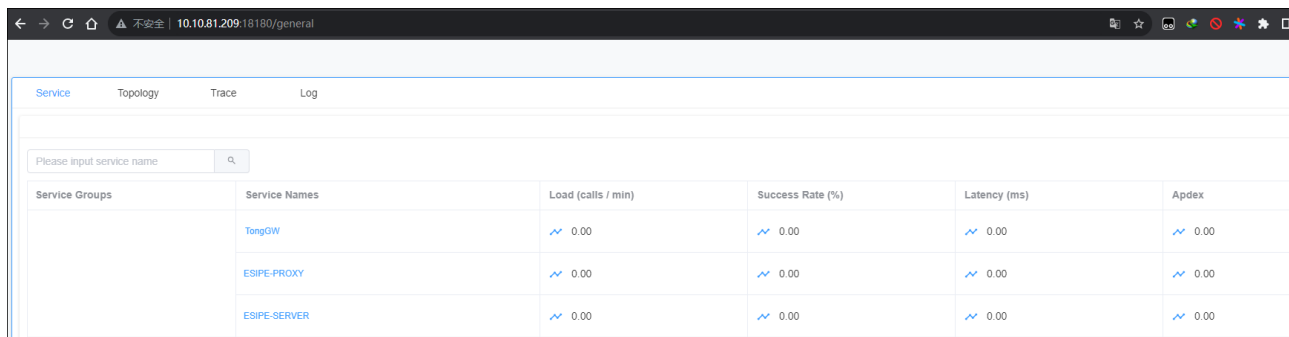
注意:

将“skywalking”集成到“管理中心”需要对“skywalking ui”进行部分定制修改。

6. 执行如下命令，重启“skywalking ui”。

```
docker restart oap_ui
```

7. 检查“skywalking”是否可正常访问。



The screenshot shows the SkyWalking UI dashboard. The top navigation bar includes 'Service', 'Topology', 'Trace', and 'Log'. Below the navigation bar is a search input field labeled 'Please input service name'. The main content area displays a table with the following columns: 'Service Groups', 'Service Names', 'Load (calls / min)', 'Success Rate (%)', 'Latency (ms)', and 'Apdex'. The table lists three service groups: 'TongGW', 'ESPIE-PROXY', and 'ESPIE-SERVER'. Each group has a corresponding row with metrics, all showing 0.00 values and a blue line icon.

Service Groups	Service Names	Load (calls / min)	Success Rate (%)	Latency (ms)	Apdex
	TongGW	0.00	0.00	0.00	0.00
	ESPIE-PROXY	0.00	0.00	0.00	0.00
	ESPIE-SERVER	0.00	0.00	0.00	0.00

1.3.3. 安装并启动 skywalking agent

本章节介绍如何安装并启动 skywalking agent。

前置条件

已获取产品工具包中的“apm-is-2.x-plugin-8.16.0.jar”。

“apm-is-2.x-plugin-8.16.0.jar”为产品提供的自定义插件。

操作步骤

1. 下载 [skywalking java agent 8.16.0](#)。
2. 解压下载的“apache-skywalking-java-agent-8.16.0.gz”。

```
tar -zxvf apache-skywalking-java-agent-8.16.0.gz
```

3. 进入解压目录“skywalking java agent”。

```
cd skywalking-agent
```

4. 添加相关插件。

- **apm-spring-cloud-gateway-2.1.x-plugin-8.16.0.jar**

将解压目录中的“/optional-plugins/apm-spring-cloud-gateway-2.1.x-plugin-8.16.0.jar”复制到解

压目录的“/plugins/”下。

```
cp ./optional-plugins/apm-spring-cloud-gateway-2.1.x-plugin-8.16.0.jar ./plugins/apm-spring-cloud-gateway-2.1.x-plugin-8.16.0.jar
```

- **apm-spring-webflux-5.x-plugin-8.16.0.jar**

将解压目录中的“/optional-plugins/apm-spring-webflux-5.x-plugin-8.16.0.jar”复制到解压目录的“/plugins/”下。

```
cp ./optional-plugins/apm-spring-webflux-5.x-plugin-8.16.0.jar ./plugins/apm-spring-webflux-5.x-plugin-8.16.0.jar
```

- **apm-is-2.x-plugin-8.16.0.jar**

注意：

“apm-is-2.x-plugin-8.16.0.jar”为自定义插件，需要手动上传至“plugins”目录。

